

General Instructions:

- The submission due date of this assignment is **Friday (3th April 2026) midnight** (before **12:00 am**)
- Write a report (i.e. in a word file) that illustrates your main solution steps including the best fitness values and the average values, plotted over generations (with and without elitism)
- Zip your code and the report in a file entitled [**YourName_YourID_AssignmentNumber**]
- This assignment should be delivered and discussed INDIVIDUALLY

Requirements:

In this assignment it is required to implement a PSO algorithm to solve the following problem:

$$\text{Max } f(x_1, x_2) = \sin(2x_1 - 0.5\pi) + 3 \cos(x_2) + 0.5x_1$$

$$\text{Where } -2 \leq x_1 \leq 3 \text{ and } -2 \leq x_2 \leq 1$$

The Algorithm for PSO can be described as follows:

1. Initialize the population of particles with random positions and velocities on D dimensions.
2. For each particle, evaluate the fitness based on the above function.
3. Find the maximum fitness & compare it with the best fitness found so far '*pbest*'. If it is better than '*pbest*', set '*pbest*' to the maximum fitness in the population and set to the location of the particle with the maximum fitness.
4. Update the velocities and the positions of the particles according to the set of equations described in the lecture.
5. Loop to step 2, until stopping criteria is met.

Plot the movement of the particles where $V_{\max} = [-0.1, 1]$

BEST OF LUCK!

